File permissions in Linux

Project description

This project demonstrates my ability to use Linux commands to manage file permissions. I analyzed existing permissions, identified incorrect access settings, and used Linux commands such as ls -la and chmod to update authorization. This activity helped me understand how security professionals manage access control in Linux systems.

Check file and directory details

```
ls -la
```

The ls -la command displays all files and directories, including hidden files, along with their permissions and ownership information

Describe the permissions string

The 10-character permissions string shows the access permissions for a file or directory. The first character represents the file type. The next three characters represent the user permissions, the next three represent the group permissions, and the last three represent permissions for others. The letters r, w, and x represent read, write, and execute permissions.]

Change file permissions

```
chmod o-w project.txt
```

The chmod o-w command removes write permissions from others for the file. This ensures that unauthorized users cannot modify the file.

Change file permissions on a hidden file

```
chmod 640 .project_x.txt
```

The chmod 640 command changes the permissions of the hidden file .project_x.txt. It gives the user read and write permissions, gives the group read permission, and removes all permissions from others.

Change directory permissions

chmod 700 drafts

The chmod 700 command changes the permissions of the drafts directory. It gives the user read, write, and execute permissions while removing all permissions from the group and others. This ensures that only the owner can access the directory and its contents.

Summary

In this project, I used Linux commands to analyze and manage file permissions. I used commands such as ls -la and chmod to review permissions, update access settings, and prevent unauthorized access to files and directories. This activity improved my understanding of how security professionals manage authorization and maintain system security in Linux environments.